

Corvinus University of Budapest

Institute of Economics

Direction is good but now is too diverse and
remains superficial.

Spatial inequality between Budapest and the countryside

Economic Complexity research proposal

- ① Engage more with known factors of regional inequality.
- ② Choose a focus: FDI(?) and planned investments(?)
migration(?)
- ③ Relate better to the course: EC, relatedness
- ④ Use more science literature than OECD-INEF

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1. Introduction

This question is very widely researched.

Spatial inequality in terms of economic output, wealth and demographics is one of the most important questions in case of any country. Hungary in particular suffers from large spatial inequality between the capital city, Budapest plus its surrounding metropolitan area and the Hungarian countryside, going back decades. With the service sector becoming more important and with the rise of the knowledge industry this unequal predicament has been accelerating since the 1990s (Brown et al, 2007). The gap is widening between Budapest and all the other regions in terms of GDP, demographics and life quality. We are interested in the question of whether this inequality is stable or has it and in which way it changed since the early 2000s.

include
Hungarian
economics
regional
science
economic
geography

The current differences between Budapest and the countryside are most pressing in case of younger cohorts of society. Young adults have a limited amount of freedom when it comes to job seeking. They find it difficult to gain high value adding jobs in the local cities or counties apart from the capital. Which implies a self-generating rising inequality. If they want to have a fitting job for their qualities, they are more likely to need to move to Budapest nowadays than ever. Furthermore, the property prices in Budapest have been rising considerably in the past decades owing partly to the influx of young people from the countryside and unequal wealth accumulation. Few opportunities at local cities and rising property prices in the capital create a very hard situation, especially for young people, but everyone else is trying to make a living, nonetheless.

decisions
How do we
know this?
use concepts
learned in
class

what are pros and cons of centralization?

If we want to prevent overcrowding in Budapest policymakers in Hungary should implement reforms to rebalance the country's economy and wealth distribution. While doing nothing, this self-generating inequality will cause future obstacles, like depopulation in small towns, full brain-drain, which will cause the countryside to fall back in progress, or even reverse the progress totally. This spatial inequality and its consequences will force qualified people to move away from their hometowns, because of the living conditions.

there is
migration
literature

The problems above are why we find it important whether a shift in Budapest's economy precedes the other regions or not and in which extent do they move together. Is it true that if Budapest becomes more prosperous the countryside follows or is it not the case? These questions are important, because if we develop Budapest further, to help the countryside we may cause just the opposite, so the policy makers must make thoughtful decisions about development, to move the country in the correct way.

long-term policy question

→ EU's cohesion vs competition
→ Hungary: Pólos program
→ smart specialization

* Attila Varga

Andres Rodriguez-Pose

Paul Krugman, Roberto Galletto

2. Literature

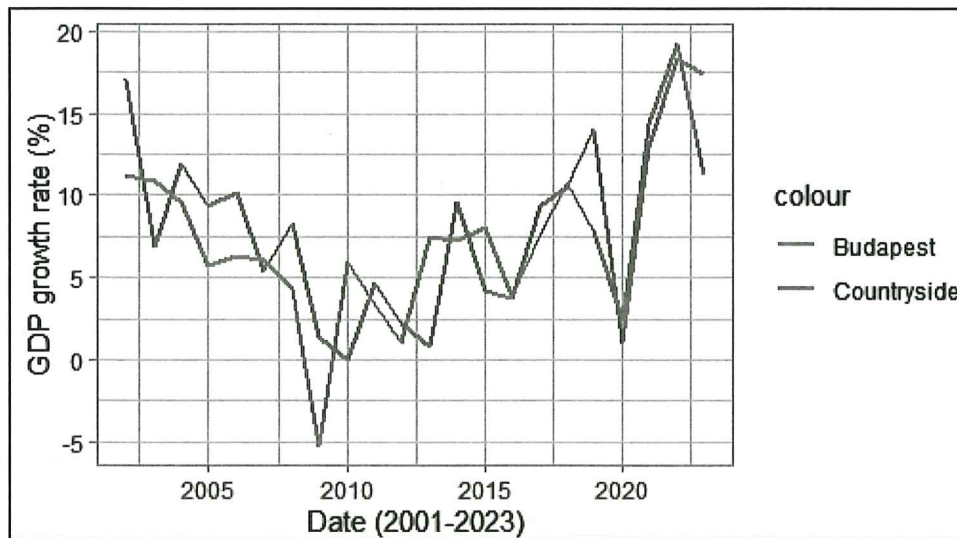
In the literature there have been a generous number of publications regarding spatial inequality, and in our case, there were some, that was conducted with Hungarian data. One of these publications is the paper of Brown et al. (2007). The authors analyzed the spatial inequality after the regime change in the 1990s, where Hungary transformed from socialism to market economy. Their scope was to identify the creation of leading sectors after the regime change, and their ability to pull other regions with themselves. They also used county level data from the Statistical Central Office, but in another timeline from 1991 to 2004. They used comparative statistics to identify these leading sectors, and the effect of FDI. They found that the difference between GDP per capita in counties didn't lower, but quite the opposite in some regions. They also showed that the quantity and the quality of foreign direct investment (FDI) contribute to the development of regions (bigger FDI, bigger growth), but the problem is that the focus of FDI was the already leading regions, where the GDP per capita was already high and where the quality of human capital was greater than in less developed regions. This implicitly assumes that this spatial inequality is a self-generating mechanism, where the divergence is going to be greater than the convergence of the regions.

In 2022 OECD published the Regions and Cities at a Glance (OECD, 2022a). This publication contains indicators on individual regions, since the new millennium. It provides an easily understandable comparison between the present and future state of nations' economic state and development. From this, we use a subsection which focuses on Hungary (OECD, 2022b). It contains a vast variety of indicators, but we only focus on the differences between Budapest and the countryside. This publication contains data on economic disparities in GDP per capita between the top 20% and the bottom 20%. The report states that: "Differences between Hungarian regions in terms of GDP per capita have decreased over the past nine years. Higher growth in the lagging regions has driven such decrease." (OECD, 2022b, p. 5). The report also contained a population projection between 2020 and 2040. Hungary's population is said to be decreasing in the upcoming two decades all around, but the biggest decrease could be seen in regions which are far from metropolitan areas, this decrease could reach 6% according to the projections. Another more detailed population growth figure showed that the from the 19 cities which are shown on the map 3 of them experienced population increase (Budapest, Sopron, Győr) and all other cities experienced decrease in population. This phenomenon shows the mobility of people in the countryside versus densely populated and developing areas.

Well FDI
se the focus?
huge effect
on FDI effect

Small with nets?
outmigration?

1. Plot: The GDP growth rate of Budapest and the Hungarian countryside (2001-2023)



source: KSH

Looking at our data, we can clearly see that there is strong correlation between the two time series. The only major difference is that Budapest never experienced a negative growth rate, not even in 2008 during the Economic Crisis or in 2020 during the Covid-19 Pandemic. Our main goal with this research is to analyze the realities of the spatially unequal Hungarian economy.

4. Methodology

Handwritten notes:

- Budapest ↔ countryside OR BP ↔ devel. ↔ less devel.

In our methodology we use a Vector Autoregression (Sims, 1980) model as the main tool to detect relations between our time series. Furthermore, we plan to measure Granger causality (Granger, 1969) to identify the existence of causality and the direction of it between past observations of one time series on the present observation of the other. For these models to work, we need to have a stationer time series. Because our independent variable is the GDP growth rate, we have an easily understandable data, which shows year by year growth in Budapest and in the countryside as well.

The conditions for a VAR model are that the time series used in it are stationer. We were curious whether our idea works or not, so we have already done some tests. First, we used the *Augmented Dickey Fuller test* (Mushtaq, 2011) to see if our time series are stationer or not. At first, the time series were not stationer, so we differentiated the data, and after that, we

The third publication that we are describing in detail is from the IMF country report, focusing on Hungary alone (IMF, 2024). The first two paragraphs clearly state that the per capita income level of Hungary is slowly converging the EU's average, meanwhile the regional disparities in Hungary are remaining high. "Budapest outpaces the rest of country while growth remains low and stagnant in the least developed southern and northern regions." (IMF, 2024, p. 1). They used beta convergence to see how much the initial GDP influences the development of counties. In this publication it was shown that at the sub national level (NUTS-3), the growth dynamics were heterogeneous between 2000-2019. Some regions (Budapest, Győr etc.) which are in the more developed parts in the country, are showing greater GDP growth, than the poorer, less developed regions.

We could see that from the early 1990s until nowadays the spatial inequality in Hungary is a big factor, that needs to be developed. The literature shows us that the more developed regions can experience greater growth, while others are falling behind. These publications can raise new questions about the sequence of growth and co-movement between Budapest and the countryside which are the main focuses of our research.

beta isigma convergence



Székely
Lengyel l.

3. Description of data collection

We collected our data from the Hungarian Central Statistics Office (KSH, 2025). We combined two data tables, one containing the GDP per capita values in HUF from 2001 to 2023 by each county plus Budapest and the other containing the population of the Hungarian counties plus Budapest for the same time period. Using these two tables we first calculated the GDP of each county and Budapest by multiplying GDP/per capita and population. We also calculated the GDP each year for the whole country as the sum of Budapest and the counties. After this we subtracted Budapest's GDP from the whole Hungarian GDP, so we got Budapest's GDP and the countryside's GDP (except the capital) for each year. As a final step we calculated the growth rate from one year to the following of Hungary except the capital (countryside) and Budapest on each observation. As a result, we got two time series, both containing 22 observations.

investments would have a rippling effect and they would automatically spread innovation and development to other areas of the country.

On the other hand, if the countryside possesses the leading role than policymakers should focus on investing in the local cities, not just the county centers, but smaller towns as well. Investing in roads, education, work efficiency measures would attract companies and foreign investment and thus it would enhance the countryside's economic potential. By developing the local regions, it would help Budapest as well, because in the long run policymakers would not have to extract as much money from there to develop other parts of the country.

If the two time series doesn't move together, the best development policy would be differential programs, where we develop the countryside in one way, and Budapest in the other, to tighten the original gap. As we mentioned in the introduction, the huge spatial inequality could lead to even larger gap, which implies reducing the growth rate of the national economy, which influences both the capital and other cities as well. In order to make correct decisions we need to know which case is true, so we need to conduct this research to get results, which can be used in the nation's way.

We saw what could happen in different cases of leading economic cycles, but how is it relevant to a country's budget and resources? Knowledge of the convergence or divergence between regions is necessary to know where to deploy our resources and where it can be used with full efficiency. The main beneficiaries from this research would be policymakers and the whole Hungarian population, because reduced inequality would possibly mean more development and higher life quality for everyone in Hungary.

Apart from ordinary people, firms would also benefit from different policies focusing on reducing inequality. Firms from the countryside would not have to locate their headquarters in Budapest because they would have the necessary consumer base and opportunities of expansion locally. Budapest-based firms would also benefit, because there would be more equal distribution of competition and costs. Foreign companies could also expand more efficiently in Hungary, because there would be other places worth investing in apart from Budapest.

conducted the ADF test again, and that resulted in stationer time series, so the models were safe to be deployed, we can conduct our research.

After the usage of the VAR model, we have to conduct several tests, to see if the residuals are white noise (meaning our model requirements are met) or not. For this, we plan to use *Breuch-Godfrey test* (Rois et al, 2012) and *Portmanteau test* (Du & Escanciano, 2012), to find out if they can be seen as white noise combined or alone or not white noise at all.

Moreover, we are going to use *Impulse-response functions* (IRF) to detect how long an impulse from one of our time series influences the other (Barnichon & Matthews, 2018) and *Forecast Error Variance Decomposition* (Gorodnichenko & Lee, 2019) to measure how our forecasts of one time series can be altered if we know the other time series.

5. Expected Outcomes

Our hypothesis regarding the outcomes is that the two time series move together and have significant influence on each other. From the fact that Budapest is Hungary's undisputed economic center we presume that Budapest's GDP growth rate will influence the whole country's economic output (meaning Budapest's GDP growth rate will be significant in the countryside's VAR model) acting as an engine for growth, spreading innovation and investment.

Maybe use networks of ~~supply~~ supply-chains, mobility, migration.

Looking at IRFs we presume that any major shock in Budapest's economy has a ripple effect on the whole country, but just in the short run. On the long run we expect the exact opposite, so any shock in the countryside's economy results in huge volatility in the case of Budapest as well. The reason behind it is that many firms have their headquarters in the capital, but their main consumer and revenue base comes from the countryside, so these would be severely impacted by a negative or positive shock coming from the counties.

6. Importance for policy makers

In this chapter the importance of the correct policy making will be described. Let's start with the situation where the economic cycles and the structural changes of Budapest predict the growth rate of the countryside. In this case, policymakers could use the signs of the economy of Budapest to predict the progress and development of the countryside. They could focus mainly on investing even more in Budapest's infrastructure and labor force, because these

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7.1. Used datasets:

KSH (2025): Population of Hungarian counties and regions on January 1st (2001 - 2025)

https://www.ksh.hu/stadat_files/nep/hu/nep0034.html

KSH (2025): GDP per capita of Hungarian counties and regions (2000-2023)

https://www.ksh.hu/stadat_files/gdp/hu/gdp0078.html